

1. Deep-Dive Mathematical Derivation from First Principles

In Nomura's Cash Equities Central Risk Book (CRB) department, we manage large institutional order flows, optimizing inventory risk and systematic asset exposures. If a candidate claims a predictive model yields an $R^2 = 1$, an Executive Director at Nomura will immediately dissect the statement from first principles.

An $R^2 = 1$ implies that the residual variance of your alpha model is exactly zero. In high-frequency equity trading or statistical arbitrage, this state represents the structural limit of perfect foresight within a specified linear subspace. Let us break down the mathematical foundations of this statement step by painstaking step.

Let y_i represent the true realized return of asset i for a given trading interval, $\hat{y}_i$ represent the conditionally predicted return generated by our alpha signals, and $\bar{y}$ represent the empirical sample mean of the realized returns across the cross-section or time-series horizon of size N .

Phase A: Line-by-Line Breakdown of the Coefficient of Determination (R^2)

The definition of the sample R^2 is written as:

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

To understand its components, we separate this fraction from first principles:

Step 1: The Residual Sum of Squares (RSS)

The numerator defines the aggregate squared deviations of true realized returns from our model's predictions:

$$\text{RSS} = \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Each individual term $e_i = y_i - \hat{y}_i$ represents the idiosyncratic residual noise or tracking error of the model.

Step 2: The Total Sum of Squares (TSS)

The denominator defines the total variance of the realized returns around their baseline sample mean:

$$\text{TSS} = \sum_{i=1}^N (y_i - \bar{y})^2$$

This acts as the variance benchmark of a naive predictor that assumes future returns will simply equal their historical cross-sectional average.

Step 3: Imposing the Boundary Condition $R^2 = 1$

We set our structural equation to 1:

$$1 - \frac{\text{RSS}}{\text{TSS}} = 1$$

Subtracting 1 from both sides of the equality yields:

$$-\frac{\text{RSS}}{\text{TSS}} = 0$$

Multiplying both sides by $-1 \cdot \text{TSS}$ (under the strict structural assumption that $\text{TSS} > 0$, meaning the asset returns are not completely static constants):

$$\text{RSS} = 0$$

Step 4: Component Vector Space Extraction

We expand the RSS summation back to its explicit vector components:

$$\sum_{i=1}^N (y_i - \hat{y}_i)^2 = 0$$

Because every single term in this summation is squared, $(y_i - \hat{y}_i)^2 \geq 0$ for all $i \in \{1, \dots, N\}$. The only mathematical way a sum of non-negative real numbers can equal exactly zero is if every individual term in the sequence is zero simultaneously:

$$(y_i - \hat{y}_i)^2 = 0 \quad \forall i$$

Taking the square root of both sides proves that:

$$y_i - \hat{y}_i = 0 \quad \forall i$$

$$y_i = \hat{y}_i \quad \forall i$$

This mathematical proof demonstrates that $R^2 = 1$ means the conditional prediction vector $\hat{\mathbf{y}}$ matches the realized outcome vector $\mathbf{y}$ across all dimensions. The model experiences zero idiosyncratic tracking variance.

Phase B: Portfolio Optimization under Zero Residual Variance

Within a Central Risk Book, we utilize mean-variance frameworks to construct optimal execution or risk-offsetting portfolios. Let $\mathbf{w} \in \mathbb{R}^{d \times 1}$ be the portfolio weight allocation vector for d cash equity positions. Let $\hat{\mathbf{r}} \in \mathbb{R}^{d \times 1}$ be our vector of predicted asset returns, and $\Sigma \in \mathbb{R}^{d \times d}$ represent the asset covariance matrix.

The standard Markowitz mean-variance optimization objective, maximizing expected return scaled by a risk-aversion parameter γ , is written as:

$$\max_{\mathbf{w}} \mathcal{J}(\mathbf{w}) = \mathbf{w}^\top \hat{\mathbf{r}} - \frac{\gamma}{2} \mathbf{w}^\top \Sigma \mathbf{w}$$

Step 1: Structural Factor Decomposition of the Covariance Matrix

Using a standard linear factor framework appropriate for a cash equities risk book, we decompose the total covariance matrix Σ into systematic factor risks and idiosyncratic residual variances:

$$\Sigma = \mathbf{B} \Omega \mathbf{B}^\top + \mathbf{D}$$

Where:

- $\mathbf{B} \in \mathbb{R}^{d \times m}$ is the factor loading matrix (e.g., exposure to Market, Style, and Sector factors).
- $\mathbf{\Omega} \in \mathbb{R}^{m \times m}$ is the systematic factor covariance matrix.
- $\mathbf{D} \in \mathbb{R}^{d \times d}$ is the diagonal matrix of specific residual asset variances: $\mathbf{D} = \text{diag}(\sigma_{\epsilon,1}^2, \sigma_{\epsilon,2}^2, \dots, \sigma_{\epsilon,d}^2)$.

Step 2: The Mathematical Consequence of $R^2 = 1$

Because our alpha prediction has an $R^2 = 1$ relative to the asset returns, there is no unmodeled or unexplained variance left in the residual term. The specific idiosyncratic variance matrix $\mathbf{D}$ collapses to the zero matrix:

$$\mathbf{D} = \mathbf{0}$$

Thus, the total asset covariance matrix simplifies entirely to its systematic projection space:

$$\mathbf{\Sigma} = \mathbf{B}\mathbf{\Omega}\mathbf{B}^\top$$

Step 3: Factor-Neutral Alpha Cross-Sectional Arbitrage

To isolate our perfect alpha predictions from systematic market movements, we enforce a strict multi-factor neutrality constraint on our allocation vector $\mathbf{w}$. We require our portfolio to have zero net exposure to all systematic risk factors:

$$\mathbf{B}^\top \mathbf{w} = \mathbf{0}$$

Let us compute the total portfolio variance under this structural risk book constraint:

$$\sigma_p^2 = \mathbf{w}^\top \mathbf{\Sigma} \mathbf{w} = \mathbf{w}^\top (\mathbf{B}\mathbf{\Omega}\mathbf{B}^\top) \mathbf{w}$$

Using the associativity property of matrix multiplication:

$$\sigma_p^2 = (\mathbf{w}^\top \mathbf{B}) \mathbf{\Omega} (\mathbf{B}^\top \mathbf{w})$$

Since $\mathbf{B}^\top \mathbf{w} = \mathbf{0}$, its transpose $\mathbf{w}^\top \mathbf{B} = \mathbf{0}^\top$. Substituting these into the portfolio variance equation yields:

$$\sigma_p^2 = \mathbf{0}^\top \mathbf{\Omega} \mathbf{0} = 0$$

Step 4: Calculating the Theoretical Portfolio Sharpe Ratio

The Sharpe Ratio (SR) of this factor-neutral portfolio is defined as its excess return divided by its total volatility:

$$\text{SR} = \frac{\mathbf{w}^\top \hat{\mathbf{r}}}{\sigma_p} = \frac{\mathbf{w}^\top \hat{\mathbf{r}}}{0} \rightarrow \infty$$

This mathematical derivation demonstrates that if $R^2 = 1$, enforcing systematic factor neutrality eliminates all portfolio variance. This creates a risk-free statistical arbitrage with an infinite theoretical Sharpe ratio.

Phase C: The Kelly Criterion and the Mathematical Necessity of Market Impact

When portfolio variance approaches zero, traditional mean-variance models dictate infinite capital allocation. To analyze this boundary condition, we can use the Kelly Criterion for optimal position sizing under leverage constraints.

Let us isolate a position in a single asset i . The optimal Kelly capital fraction w_i^* to allocate is proportional to the expected return divided by the variance of the return:

$$w_i^* = \frac{\mathbb{E}[r_i]}{\sigma_i^2}$$

As proven in Phase B, if $R^2 = 1$ and systematic risks are neutralized, the conditional variance $\sigma_i^2 \rightarrow 0$. Evaluating this limit yields:

$$\lim_{\sigma_i^2 \rightarrow 0} w_i^* = \lim_{\sigma_i^2 \rightarrow 0} \frac{\mathbb{E}[r_i]}{\sigma_i^2} = \infty$$

In an unconstrained market, the model demands infinite leverage. In a real institutional setting like Nomura, this boundary condition is constrained by two microstructural factors: **Transaction Costs** and **Market Impact**.

Step 1: Quadratic Market Impact Modeling

When executing large block volumes within a CRB, buying or selling assets shifts the market clearing price against us. We model the realized return net of execution costs using a quadratic market impact function:

$$r_i^{\text{realized}}(q_i) = \hat{r}_i - \theta_i q_i$$

Where:

- q_i is the absolute quantity or size of shares executed by our book.
- $\hat{r}_i$ is the true, unimpacted fair alpha prediction ($R^2 = 1$).
- $\theta_i > 0$ represents the microstructural instantaneous market impact coefficient for asset i .

Step 2: Formulating the Net PnL Objective Function

The net expected PnL of our position as a function of trade size q_i is written as:

$$\mathbb{E}[\text{PnL}(q_i)] = q_i \cdot r_i^{\text{realized}}(q_i) = q_i (\hat{r}_i - \theta_i q_i)$$

Distributing q_i across the function yields a non-linear objective:

$$\mathbb{E}[\text{PnL}(q_i)] = \hat{r}_i q_i - \theta_i q_i^2$$

Step 3: Optimization via First-Order Conditions

To maximize net expected PnL under perfect predictive foresight, we take the derivative of our PnL function with respect to trade size q_i and set it to 0:

$$\frac{d\mathbb{E}[\text{PnL}(q_i)]}{dq_i} = \hat{r}_i - 2\theta_i q_i = 0$$

$$\hat{r}_i = 2\theta_i q_i$$

Solving for the optimal allocation size q_i^* :

$$q_i^* = \frac{\hat{r}_i}{2\theta_i}$$

Step 4: Maximum Attainable Bounded Return

We substitute the optimal trade size q_i^* back into our net PnL expression to find our maximum achievable returns:

$$\mathbb{E}[\text{PnL}(q_i^*)] = \hat{r}_i \left(\frac{\hat{r}_i}{2\theta_i} \right) - \theta_i \left(\frac{\hat{r}_i}{2\theta_i} \right)^2$$

$$\mathbb{E}[\text{PnL}(q_i^*)] = \frac{\hat{r}_i^2}{2\theta_i} - \theta_i \left(\frac{\hat{r}_i^2}{4\theta_i^2} \right)$$

Canceling out the impact parameter θ_i in the second term:

$$\mathbb{E}[\text{PnL}(q_i^*)] = \frac{\hat{r}_i^2}{2\theta_i} - \frac{\hat{r}_i^2}{4\theta_i} = \frac{2\hat{r}_i^2 - \hat{r}_i^2}{4\theta_i}$$

$$\mathbb{E}[\text{PnL}(q_i^*)] = \frac{\hat{r}_i^2}{4\theta_i}$$

This proof demonstrates that even with a perfect alpha model ($R^2 = 1$), market impact prevents infinite leverage and caps the book's expected returns. The maximum capacity of your strategy is structurally constrained by liquidity, scaling inversely with the market impact coefficient θ_i .

2. Quantitative Infrastructure & Application of Theory

The following production-grade script provides a full implementation of an institutional optimization and simulation engine for a central risk book. It simulates cash equity assets driven by structural risk factors, applies a high- R^2 alpha predictive signal, constructs factor-neutral weights, handles quadratic market impact limits, and renders a comprehensive validation dashboard using Plotly.

```
import numpy as np
import pandas as pd
import plotly.graph_objects as go
from plotly.subplots import make_subplots

# =====
# 1. STRUCTURAL CRB MARKET & ALPHA DATAFRAME GENERATION
# =====
def generate_crb_alpha_data(n_assets: int = 50, seed: int = 42) -> tuple:
    """
    Generates systematic cash equity factor exposures and an alpha signal matrix.
    Includes asset parameter maps for analyzing market impact limits.
    """
    np.random.seed(seed)

    # Define asset tickers for our index universe
    tickers = [f"STK_{i:03d}" for i in range(n_assets)]

    # Simulate systematic factor loadings (e.g., Market Beta, Sector
    # Sensitivities)
    market_betas = np.random.uniform(0.5, 1.5, size=n_assets)
```

```

sector_loadings = np.random.uniform(-0.8, 0.8, size=n_assets)

# Generate the unimpacted raw alpha signal vector (perfect prediction
benchmark)
true_alpha_signals = np.random.normal(loc=0.0015, scale=0.004, size=n_assets)

# Define microstructural market impact coefficients (theta_i)
# Less liquid assets have higher theta values
market_impact_thetas = np.random.uniform(1e-5, 8e-5, size=n_assets)

df = pd.DataFrame({
    'Asset_Ticker': tickers,
    'Market_Beta': market_betas,
    'Sector_Loading': sector_loadings,
    'Raw_Alpha_Signal': true_alpha_signals,
    'Impact_Theta': market_impact_thetas
}).set_index('Asset_Ticker')

# Construct the structural systematic design constraint matrix B (N x 2)
B_matrix = df[['Market_Beta', 'Sector_Loading']].values

return df, B_matrix

# =====
# 2. FACTOR-NEUTRAL OPTIMIZATION ENGINE WITH IMPACT CAPABILITIES
# =====
class CRBOptimizationEngine:
    """
    Solves for optimal trade sizing under systematic factor-neutral constraints
    and quadratic market impact limits.
    """
    def __init__(self, asset_df: pd.DataFrame, B_matrix: np.ndarray):
        self.df = asset_df
        self.B = B_matrix
        self.N = len(asset_df)

    def compute_naive_allocations(self) -> np.ndarray:
        """Allocates capital proportionally to signals, ignoring risk
constraints."""
        signals = self.df['Raw_Alpha_Signal'].values
        return signals / np.sum(np.abs(signals))

    def compute_impact_optimal_allocations(self) -> np.ndarray:
        """
        Computes the analytical capacity cap allocation derived in Phase C:
 $q_i = \alpha_i / (2 * \theta_i)$ 
        """
        alphas = self.df['Raw_Alpha_Signal'].values
        thetas = self.df['Impact_Theta'].values
        return alphas / (2.0 * thetas)

    def enforce_factor_neutrality(self, target_weights: np.ndarray) -> np.ndarray:
        """
        Projects allocations into the null space of the systematic factor matrix B

```

```

    to ensure strict factor neutrality:  $B^T * w = 0$ .
    """
    # Compute the projection matrix into the null space of B
    #  $P = I - B * (B^T * B)^{-1} * B^T$ 
    I = np.eye(self.N)
    Bt_B_inv = np.linalg.inv(self.B.T @ self.B)
    projection_matrix = I - self.B @ Bt_B_inv @ self.B.T

    neutral_weights = projection_matrix @ target_weights
    return neutral_weights

# =====
# 3. HIGH-FIDELITY PLOTLY INTERACTIVE DIAGNOSTIC CANVASES
# =====
def generate_crb_diagnostic_dashboard(df: pd.DataFrame, w_naive: np.ndarray,
w_opt_neutral: np.ndarray):
    """
    Generates an interactive dashboard to evaluate the performance of our risk
    book model,
    comparing unconstrained allocations against our factor-neutral, impact-
    adjusted weights.
    """
    fig = make_subplots(
        rows=2, cols=2,
        subplot_titles=(
            "Cross-Sectional Trade Sizes: Naive vs Risk-Optimized",
            "Systematic Factor Exposure Residual Profiles",
            "Realized Net Return Curves Under Market Impact Capacity Limits",
            "Total Expected Strategy Capacity Profile"
        ),
        vertical_spacing=0.15, horizontal_spacing=0.12
    )

    asset_indices = np.arange(len(df))

    # Panel 1: Bar Chart of Allocation Sizes
    fig.add_trace(go.Bar(x=asset_indices, y=w_naive * 1e4, name='Naive Allocation
(bps)', marker_color='crimson'), row=1, col=1)
    fig.add_trace(go.Bar(x=asset_indices, y=w_opt_neutral, name='Optimized Sizes
(Shares)', marker_color='emerald'), row=1, col=1)

    # Panel 2: Factor Exposure Residual Profiles (Checking  $B^T * w = 0$ )
    B_market = df['Market_Beta'].values
    B_sector = df['Sector>Loading'].values

    naive_market_exp = np.sum(w_naive * B_market)
    naive_sector_exp = np.sum(w_naive * B_sector)
    opt_market_exp = np.sum(w_opt_neutral * B_market)
    opt_sector_exp = np.sum(w_opt_neutral * B_sector)

    fig.add_trace(go.Bar(
        x=['Market Beta (Naive)', 'Sector Load (Naive)', 'Market Beta
(Optimized)', 'Sector Load (Optimized)'],
        y=[naive_market_exp, naive_sector_exp, opt_market_exp, opt_sector_exp],

```

```

        marker_color=['indianred', 'indianred', 'mediumseagreen',
'mediumseagreen'],
        name='Factor Exposures'
    ), row=1, col=2)

# Panel 3: Net Return Curves as a Function of Trade Volume (Asset 0)
alpha_0 = df['Raw_Alpha_Signal'].values
theta_0 = df['Impact_Theta'].values
q_range = np.linspace(0, alpha_0 / theta_0, 100)

gross_returns = alpha_0 * q_range
net_pnl_curve = alpha_0 * q_range - theta_0 * (q_range ** 2)

fig.add_trace(go.Scatter(x=q_range, y=gross_returns, mode='lines',
line=dict(color='cyan', width=2, dash='dash'), name='Gross Linear Return'), row=2,
col=1)
fig.add_trace(go.Scatter(x=q_range, y=net_pnl_curve, mode='lines',
line=dict(color='gold', width=3), name='Net PnL (Impact Constrained)'), row=2,
col=1)
# Add marker at the theoretical peak verified in Phase C: q = alpha / 2*theta
fig.add_trace(go.Scatter(x=[alpha_0 / (2*theta_0)], y=
[(alpha_0**2)/(4*theta_0)], mode='markers', marker=dict(size=12, color='orange',
symbol='star'), name='Optimal Capacity Peak'), row=2, col=1)

# Panel 4: Expected Net PnL Breakdown across the asset cross-section
net_pnl_optimized_cross_section = (df['Raw_Alpha_Signal'].values ** 2) / (4.0
* df['Impact_Theta'].values)
fig.add_trace(go.Bar(x=asset_indices, y=net_pnl_optimized_cross_section,
marker_color='royalblue', name='Net PnL Capacity per Asset'), row=2, col=2)

fig.update_layout(
    title_text="Nomura Cash Equities CRB – Efficient Portfolio Frontier under
Perfect Prediction ( $R^2 = 1\$$ )",
    title_x=0.5, height=900, width=1300, template="plotly_dark",
    showlegend=False
)

fig.update_xaxes(title_text="Asset Cross-Sectional Indexes", row=1, col=1)
fig.update_yaxes(title_text="Allocation Scale Metric", row=1, col=1)
fig.update_yaxes(title_text="Net Systematic Factor Sensitivity Exposure",
row=1, col=2)
fig.update_xaxes(title_text="Executed Share Quantity ( $q_i$ )", row=2, col=1)
fig.update_yaxes(title_text="Expected Return / Strategy PnL Score", row=2,
col=1)
fig.update_xaxes(title_text="Asset Cross-Sectional Indexes", row=2, col=2)
fig.update_yaxes(title_text="Max Attainable Return [ $\alpha^2 / 4*\theta$ ]",
row=2, col=2)

fig.show()

# =====
# 4. ENGINE EXECUTION GATEWAY CONTROL ENTRY POINT
# =====
if __name__ == "__main__":

```



```

# Initialize 50 assets inside our central risk book environment
crb_df, structural_B_matrix = generate_crb_alpha_data(n_assets=50, seed=101)

# Initialize our optimization engine
book_optimizer = CRBOptimizationEngine(crb_df, structural_B_matrix)

# Compute baseline allocations and optimal sizes
naive_w = book_optimizer.compute_naive_allocations()
optimal_q = book_optimizer.compute_impact_optimal_allocations()

# Project optimal sizes into the factor null space to ensure strict factor
neutrality
neutral_optimal_q = book_optimizer.enforce_factor_neutrality(optimal_q)

print("\n" + "="*65)
print("NOMURA CRB PORTFOLIO CALIBRATION REPORT:")
print("="*65)
print(f"Total Portfolio Asset Count      : {len(crb_df)}")
print(f"Naive Strategy Market Exposure : {np.sum(naive_w *
crb_df['Market_Beta'].values):.6f}")
print(f"Naive Strategy Sector Exposure : {np.sum(naive_w *
crb_df['Sector_Loading'].values):.6f}")
print("-"*65)
print(f"Optimized Market Risk Exposure : {np.sum(neutral_optimal_q *
crb_df['Market_Beta'].values):.6f} (Target: 0.00)")
print(f"Optimized Sector Risk Exposure : {np.sum(neutral_optimal_q *
crb_df['Sector_Loading'].values):.6f} (Target: 0.00)")
print("-"*65)

total_net_pnl_capacity = np.sum((crb_df['Raw_Alpha_Signal'].values ** 2) /
(4.0 * crb_df['Impact_Theta'].values))
print(f"Total Strategy PnL Capacity Cap : {total_net_pnl_capacity:.4f}
currency units")
print("="*65 + "\n")

# Generate dashboard
generate_crb_diagnostic_dashboard(crb_df, naive_w, neutral_optimal_q)

```

3. Strategic Interview Follow-Ups & Edge-Case Vulnerabilities

Follow-Up 1: Model Overfitting and Out-of-Sample Collapse

Question: "Suppose you train an in-sample regression model that achieves a perfect $R^2 = 1$. When deployed live in our Cash Equities Central Risk Book, the out-of-sample R^2 collapses to -0.45 . What structural errors occurred during your modeling process, and what are the risk implications for the book's capital?"

- **The Answer:** Achieving an in-sample $R^2 = 1$ that collapses out-of-sample is a classic symptom of severe **overfitting** or data leakage. This occurs when the model has too many parameters relative to

the number of historical observations ($d \geq N$), allowing it to interpolate training noise perfectly rather than learning true systematic relationships.

- **The Mathematical Defect:** When $d \geq N$, the design matrix $\mathbf{F}^\top \mathbf{F}$ becomes rank-deficient, and the model's residual vector is driven to zero by construction. Out-of-sample, the model fails to generalize, and its predictions $\hat{y}_i$ can diverge significantly from realized returns. This results in a Residual Sum of Squares (RSS) that exceeds the Total Sum of Squares (TSS), driving the out-of-sample R^2 below zero:

$$\text{RSS}_{\text{OOS}} > \text{TSS}_{\text{OOS}} \implies R_{\text{OOS}}^2 = 1 - \frac{\text{RSS}_{\text{OOS}}}{\text{TSS}_{\text{OOS}}} < 0$$

- **The Risk to the Book:** Because the optimization engine assumed the model's residual variance was zero, it did not apply any idiosyncratic risk constraints ($\mathbf{D} = \mathbf{0}$). When deployed live, the book will take on massive, unhedged idiosyncratic positions. As the out-of-sample model fails, these positions will suffer significant losses, leading to severe drawdown risk.

Follow-Up 2: Non-Linear Market Impact and Multi-Asset Cross-Impact

Question: "In your derivation, you assumed a linear marginal market impact function ($\theta_i q_i$), which leads to a quadratic cost term. However, large liquidations typically exhibit a square-root market impact function ($\theta_i \sqrt{q_i}$). Additionally, liquidating a large block in asset i can impact the liquidity and price of a correlated asset j via cross-impact vectors. How do these dynamics alter your optimal capacity equation?"

- **The Answer:** Moving from a linear to a square-root market impact function changes the shape of our net PnL objective. Incorporating cross-impact dynamics replaces our asset-by-asset optimization with a coupled system across the entire portfolio.
- **The Non-Linear Mathematical Framework:** Let $\mathbf{q} \in \mathbb{R}^{d \times 1}$ represent our executed volume vector. We model the multi-asset price impact vector using a non-linear cross-impact matrix function:

$$\Delta \mathbf{r}(\mathbf{q}) = \Theta \cdot \text{sign}(\mathbf{q}) \odot |\mathbf{q}|^\alpha$$

Where $\Theta \in \mathbb{R}^{d \times d}$ is a non-diagonal cross-impact matrix, $\odot$ represents the Hadamard element-wise product, and $\alpha \approx 0.5$ captures the empirical square-root scaling rule.

- **The Optimization Challenge:** The net portfolio return objective becomes:

$$\mathbb{E}[\text{PnL}(\mathbf{q})] = \mathbf{q}^\top \hat{\mathbf{r}} - \mathbf{q}^\top (\Theta (\text{sign}(\mathbf{q}) \odot |\mathbf{q}|^\alpha))$$

- Because $\alpha = 0.5$, the objective function is non-quadratic, and its derivative cannot be solved using simple linear algebra. Additionally, because Θ contains non-zero off-diagonal elements ($\theta_{i,j} \neq 0$), a liquidation in stock i will generate execution costs and price shifts in stock j .
- To solve for the optimal trade sizes $\mathbf{q}^*$ under these realistic conditions, the central risk book must replace its analytical models with numerical optimization frameworks, such as sequential quadratic programming (SQP) or interior point methods.